

7/26/23, 1:26 AM

Codio - Distributing with Docker and Heroku

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TerminalDockerfilex1 FROM continuumio/anaconda3:2021.1123 ADD . /code45 WORKDIR /code67 ENTRYPOINT ["python", "birthday/main.py"]

GuidexCollapse

Dockerfile

Dockerfile, Image, and Container

Start the `birthday` virtual environment.

```
conda activate birthday
```

We previously talked about the advantages using a container has over using a virtual machine. We will be using the Docker container system to distribute our Flask app. First, let's start with some vocabulary. When we say our app is "running in Docker", what we really mean is the **container** is running an instance of a Docker image.

A **Docker image** is the environment that runs our application. We can select things like the operating system, the version of Python, any dependencies for our app, etc. You can be very selective about what is and is not a part of your Docker image. For example, Alpine Linux is a popular choice as the foundation of a Docker image. This OS is tiny and consumes hardly any resources. This helps with performance. Also, the fewer packages that are installed on the image, the fewer potential security holes.

The instructions for building a Docker image are contained in the **Dockerfile**. This is a blueprint that builds the environment that runs our app step by step. Every time you want to use Docker for a project, you need to start with a Dockerfile. Every Dockerfile sets the foundation with an image, then we customize it for our needs. This could be installing additional software, copying files, running commands, etc. Start by creating the file `Dockerfile` itself. Then use the link below to open it.

```
touch Dockerfile
```

▼ Did you notice?

The name for `Dockerfile` starts with a capital "D". This is a requirement. The file does not have an extension at the end either.

Open Dockerfile

We are going to keep things very simple with our Dockerfile. First, we are going to use the latest version (2021.11 at the time of writing) of `anaconda3` as the foundation of our project. Anaconda maintains the `anacoda3` image. They provide us with a pre-made selection of software. In particular, it gives us access to Python and Flask. This means we do not have to tell our Dockerfile to install this dependency. You could argue using the `anaconda3` image is excessive given our needs.

```
FROM continuumio/anaconda3:2021.11
```

Next, we need to copy any relevant files to the Docker image. We want all of our files, which is represented by the `.`. Again, we could probably leave off some files, but we are keeping things simple. Copy our files to `/code` directory on the Docker image. Then set our working directory to `/code`.

```
FROM continuumio/anaconda3:2021.11

ADD . /code

WORKDIR /code
```

Finally, our entry point serves as the command that starts up our Flask app. The first element in the list is the command to run a Python script. Remember, we built our Flask app inside the `birthday` directory. So use `birthday/main.py` as the path for our script.

```
FROM continuumio/anaconda3:2021.11

ADD . /code

WORKDIR /code

ENTRYPOINT ["python", "birthday/main.py"]
```

That is all their is to our simple Dockerfile. When we build our custom Docker image we start with an image from Anaconda. This has several packages already installed, most importantly Flask. We then copy our work into a directory on Docker called `/code`. This directory is set as our working directory. Finally, we give the command and file path to run our Flask app.

In the terminal, deactivate the `birthday` virtual environment.

```
conda deactivate
```

Reading Question

Select the name of the file used to create a Docker image.

☐ Dockerfile.txt

☒ Dockerfile

☐ dockerfile

☐ Docker.file

The file used to create a Docker image is called "Dockerfile". This file must start with a capial "D" and does not have a file extension.

Check It!

Next

100% (7:42)

https://codio.com/sandipan-dey/distributing-with-docker-and-heroku/tree/Dockerfile

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